Impact of Ship-Generated Waves on the Sediment Resuspension in Restricted Waterways

Mainak Chakraborty[1]; V. Sriram[2]; and K. Murali[3]

Abstract: The ship-generated waves in restricted waterways can influence the behavior of sediment resuspension and bank erosion. This paper explores the case study considering the effect of ship-generated waves in restricted channels. A fully nonlinear Boussinesq model (FUNWAVE-TVD) is used to simulate the ship-generated waves. The sediment transport model adopted in FUNWAVE has been changed to incorporate the modified diffusion coefficients, deposition flux, and an explicit equation for settling velocity and critical shield parameter. The sediment transport model was validated against the experimental results of solitary waves in a wave tank, and model performance in shoreline geometry was reported. The implementation of the moving vessel in the numerical model and its impact on suspended sediment concentration were evaluated against the field measurements in the Falta stretch of the Hooghly river in Indian National Waterway 1. The error value in predicting the depth-averaged suspended sediment concentration by the FUNWAVE module was 97.6%, which has been significantly reduced to 16.3% based on the present modification. The parametric study of the ship waves in a typical restricted waterway of India revealed that the concentration of the sediments gradually decreases from the navigation channel to the river banks due to dissipation in the ship waves. The change in wave height and depth-averaged sediment concentration increases with the increasing depth Froude number. Finally, the impact on sediment transport due to different scenarios, such as the movement of two vessels in close proximity, the passage of vehicles coming from the opposite direction on the navigation channel, the influence of small draft vessels, and the bed morphology changes due to the multiple vessel movements are analyzed and reported. **DOI: 10.1061/JWPED5.WWENG-1951**. *© 2023 American Society of Civil Engineers*.

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Introduction

Inland waterways are the most ecofriendly, sustainable, and economical mode of transportation due to less consumption of energy and carbon dioxide emission than land transportation. However, increasing human interference in inland waterways has negatively impacted the marine ecosystem and is a matter of concern worldwide. Usually, waves are the major driving force for the littoral transport of sediments on both high-energy shorelines (Abadie et al. 2006) and on low-energy shores of sheltered sea coasts or lakes (Hofmann et al. 2008). Wind, tidal fluctuations, and earthquakes are the major sources of wave generation in coastal and inland waterways. In the recent past, waves generated by marine vehicles have been of utmost importance to researchers, as the ecosystem of inland waterways may be destroyed by the increasing movement of marine vehicles, which introduce complex hydrodynamic forcing in combination with wind and tides. Shorelines are important natural infrastructures along our coasts since they provide ecosystem services in the form of erosion mitigation, biological diversity, and recreational and aesthetic values (Barbier et al. 2011). The major impact of the ship-generated waves has been witnessed in the river bank and shoal areas due to the wave-breaking process (Fleit et al. 2019). Ship-generated waves can contribute to shoreline erosion (Sorensen 1997; Bourne 2000; Gaskin et al. 2003; Nanson et al. 1994) and resuspension of bottom materials, loss of biological habitat, aquatic wildlife (Ali et al. 1999; Lindholm et al. 2001), trigger ecological imbalance, reduce biodiversity (Gabel et al. 2017), and decrease the perceived aesthetic values of the shore. Thus, to protect shorelines in navigational fairways, it is necessary to consider the hydrodynamic forces and sediment resuspension induced by ships (Roo and Troch 2013).

Ship movement in waterways generates water surface fluctuations separated into two systems consisting of primary and secondary waves (Bertram 2012; Havelock 1908). While the study of the ship waves in the deep waters has been well reported and documented in a generalized manner (Havelock 1908; Thomson 1887; Sorensen and Weggel 1984), the study of ship-generated waves in shallow regions are few due to their complex nature. They are governed mainly by local hydrodynamic flow conditions, channel bathymetry, and limited river width. The water displacement caused by vessel navigation in shallow waters forms a depression in the water surface and an overall increase in velocity around the ship (Gourlay 2008). The increase in the water velocity causes a subsequent pressure drop leading to a depression in the midship (Sorensen 1997). This feature causes the depression wave (Soomere 2007) (alternatively called a Bernoulli wave), which can propagate as a highly nonlinear, asymmetric wave onto adjacent shallow-water areas (Parnell et al. 2015). The second set of waves that predominately forms in the wake path of ship navigation is a classical Kelvin wake wave, characterized by transverse and divergent waves (Soomere et al. 2009). This entire ship-induced

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disturbance system is regarded as the secondary wave system (Bertram 2012).

As the ship waves propagate toward the riverbanks, they shoal and waves eventually break. The shallow-water flows cause high bed shear stress and resuspension of fine sediment particles. The resuspended sediment particles are transported with the river currents and deposited downstream, thus increasing the overall dredging cost of the waterways (Gelinas et al. 2013; Houser 2010; Parnell et al. 2008; Rapaglia et al. 2011; Schoellhamer 1996; Soomere 2006; Soomere et al. 2009). In contrast to the slowly developing wind waves, the ship-generated waves cause instantaneous resuspension of sediments due to a sudden increase in water velocity within a few seconds (Rapaglia et al. 2015). It is worth mentioning that the ship-generated waves contribute more to the sediment resuspension than wind waves in fetch-restricted waterways.

The introduction of deep-draft vessels in inland waterways and the implementation of new plans related to cargo shipping in fairways have posed a major threat to the overall inland waterway stability (Uliczka and Kondziella 2016). Deep-draft vessels at shallow regions may generate fundamentally different wave systems than conventional small vessels (Brown et al. 1989; Chen and Sharma 1995; Li and Sclavounos 2002; Schoellhamer 1996). Ship-generated solitons usually have less height (Whittaker et al. 2001) and may not be an immediate threat to the people or beach environment. However, the superposition of two solitonic waves generated by two deep-draft vessels in a shallow environment may amplify water surface elevation (Peterson et al. 2003). The combination of a deeper channel and larger ships would increase the size of the divergent and transverse waves along the river (Maynord 2007). Many researchers investigated the vessel movement's hydrodynamic aspect, which was reported in the literature. Shi et al. (2018) studied the breaking of ship bores by using nonlinear Boussinesq modeling (FUNWAVE-TVD). David et al. (2017) have used the Boussinesq-type model (BOSZ) in the port of Hamburg. Ji et al. (2014) investigated the ship-generated hydrodynamics in restricted waterways and reproduced the primary and secondary waves. However, the impact of ship-generated waves on sediment resuspension is not well documented. Few studies have been performed on the effect of the multiple vessel movement on the sediment resuspension in the restricted waterways (El Safty and Marsooli 2020), and there is a need for further analysis. The impact of the small boats and barges with respect to the large deep-draft vessels has not been studied previously. Further, the correlation between the depth Froude number and suspended sediment concentration has not been explored. Therefore, the paper's main objective is to study the sediment resuspension and bed changes in restricted waterways due to ship-generated hydrodynamics by using fully nonlinear Boussinesq modeling (FUNWAVE-TVD).

The main motivation behind the modification in the sediment transport model is that FUNWAVE uses the diffusion coefficient given by Elder (1959), which was based on well-mixed shallow water flows and pollutant mixing. So, it is inappropriate for a ship wake study. Further, in the existing FUNWAVE model, the user has to provide the fall velocity as well as there is no provision to provide the background concentration on the whole computational domain. However, it is very difficult to measure fall velocity at site conditions. Therefore, the authors tried to address this by adopting different formulations for diffusion coefficients, deposition flux, and an explicit equation for settling velocity and critical shield parameter.

The present paper is structured as follows. The governing equations for the numerical models are discussed. The validation of the hydrodynamic and sediment transport model from the experimental results of Kobayashi and Lawrence (2004) is reported. The implementation of the moving vessel in the numerical model and its impact on suspended sediment concentration is evaluated against the field measurements in the Falta stretch of the Hooghly river in Indian National Waterway 1 in the subsequent section. Then, the paper discusses different aspects of restricted waterways due to vessel movements, such as the resuspension of sediments and the correlation of depth Froude number with wave height and sediment concentration. Finally, the different possible scenarios, such as the impact on the sediment transport due to the nearby movement of two vessels, erosion of the navigation bed due to vessel movement in the same path from the opposite direction, and the overall change in bed morphology due to the passage of several small draft vessels in comparison with the large draft vessels, are investigated.

Methodology

As the inland waterways are shallow waters, the effect of the vertical motion is insignificant. Therefore, depth-averaged two-dimensional (2D) equations generally cater to the hydrodynamics in restricted waterways (Sabbagh-Yazdi et al. 2007). The present paper uses the total variation diminishing (TVD) version of the fully nonlinear Boussinesq wave model (FUNWAVE) (Shi et al. 2012). A modified sediment transport model has been used for morphology changes. The description of the numerical model is now presented.

Hydrodynamic Model

FUNWAVE-TVD (Shi et al. 2012) is an open-source high-order adaptive time-stepping solver for the fully nonlinear Boussinesq model, which includes the eddy viscosity breaking module (Shi et al. 2018). Shi et al. (2012) followed the works of Chen (2006) to obtain the model equation for FUNWAVE-TVD. The complete model documentation and numerical schemes are reported by Shi et al. (2012). To simulate the ship waves, the hydrodynamic model solves the hybrid finite volume and finite difference scheme for fully nonlinear Boussinesq equations with the inclusion of a pressure source term according to Ertekin et al. (1986), Wu (1987), or Torsvik et al. (2008). The list of equations for the present work is mentioned in this section.

The governing mass conservation equation is given as

$$\eta_t + \nabla \cdot \mathbf{M} = 0 \tag{1}$$

where t = time; η = water surface elevation; and the horizontal volume flux $\mathbf{M}$ is given by

$$\mathbf{M} = H(\mathbf{u_\alpha} + \overline{\mathbf{u_2}}) \tag{2}$$

where the total water depth $H = h + \eta$ and h is the still water depth. Here, $\boldsymbol{u_\alpha}$ is the velocity at the reference elevation and $\overline{\boldsymbol{u_2}}$ denotes the depth-averaged contribution of horizontal velocity.

The depth-averaged momentum equation is given by

$$M_t + \nabla \cdot \left[\frac{\mathbf{MM}}{H}\right] + \nabla\left[\frac{1}{2}g(\eta^2 + 2h\eta)\right] = \mathbf{V}_{\text{dis}} + g\eta\nabla h - S_{\text{ship}} - S_{brk} \tag{3}$$

where g is the acceleration due to gravity, $\mathbf{V}_{\text{dis}}$ is the Boussinesq dispersive terms, S_{brk} is the dissipative term for the eddy viscosity, and S_{ship} is the pressure source term for the generation of ship wakes, given by the expression $gH\nabla p_a$. The pressure disturbance with a center point at (x^*, y^*) inside the rectangle of length L

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and width R is given by

$$p_a(\tilde{x}, \tilde{y}) = Pf(\tilde{x}, t)q(\tilde{y}, t) \tag{4}$$

$$f(\tilde{x}, t) = \begin{cases} \cos^2\left[\frac{\pi(\tilde{x}-x^*(t)-\frac{1}{2}\alpha_1 L}{(1-\alpha_1)L}\right], & \frac{1}{2}\alpha_1 L < \tilde{x} - x^*(t) \le \frac{1}{2}L \\ \cos^2\left[\frac{\pi(x^*(t)-\tilde{x}-\frac{1}{2}\alpha_2 L}{(1-\alpha_2)L}\right], & -\frac{1}{2}L \le \tilde{x} - x^*(t) < -\frac{1}{2}\alpha_2 L \\ 1, & -\frac{1}{2}\alpha_2 L \le \tilde{x} - x^*(t) \le \frac{1}{2}\alpha_1 L \end{cases} \tag{5}$$

$$q(\tilde{y}, t) = \begin{cases} \cos^2\left[\frac{\pi(|\tilde{y}-y^*(t)|-\frac{1}{2}\beta R}{(1-\beta)R}\right], & \frac{1}{2}\beta R < |\tilde{y} - y^*(t)| \le \frac{1}{2}R \\ 1, & |\tilde{y} - y^*(t)| \le \frac{1}{2}\beta R \end{cases} \tag{6}$$

where the pressure outside the region is zero, p_a is the static depression around the ship, and $(\tilde{x}, \tilde{y})$ is the pressure disturbance coordinate system that can be rotated by an angle relative to the Boussinesq coordinate system (x, y). In addition, α_1, α_2, and β are the parameters governing the shape of the vessel draft.

Sediment Transport Model

The sediment transport module of the FUNWAVE uses a depth-averaged sediment concentration equation derived from the advection–diffusion equation, which is given by

$$(\overline{c}H)_t + \nabla_{\mathbf{h}} \cdot (\overline{c}H(u_\alpha + \overline{u_2})) = \nabla_{\mathbf{h}} \cdot (kH(\nabla_h \overline{c})) + E - D \tag{7}$$

The nondimensional depth-averaged sediment concentration with sediment density is given as $\overline{c}$ and the horizontal diffusion coefficient k calculated by the formula (Elder 1959)

$$k = 5.93 u_{*c} H \tag{8}$$

whereas u_{*c} is the shear velocity and E is the erosion rate calculated by Rijn and Leo (1984).

In the present study, different coefficients for k and D have been adopted. The present modification produced a better comparison in the benchmark-validated case reported in the subsequent sections.

The diffusivity coefficients are related to the turbulent eddy viscosity (Cheng et al. 2014) such as $k_x = k_y = \nu_t/\sigma_t$, where σ_t is the turbulent Prandtl–Schmidt number and ν_t is the turbulent eddy viscosity (Rijn and Leo 1984; Sabbagh-Yazdi et al. 2007).

Here, E and D are the erosion and deposition rates given as

$$E = w_s C_a \tag{9}$$

$$D = w_s C \tag{10}$$

where w_s is the sediment fall velocity given as (Soulsby 1997)

$$w_s = \frac{\nu}{d_{50}}[(10.36^2 + 1.049 D_*^3])^{1/2} - 10.36] \tag{11}$$

D_* is the dimensionless grain size given as

$$D_* = \left[\frac{g(s-1)}{\nu^2}\right]^{1/3} d_{50} \tag{12}$$

d_{50} is the mean grain size diameter of the sediment, s is the specific gravity of the sediments, and ν is the kinematic viscosity of water.

Here, C_a is the sediment concentration close to the bed. The parameters influencing the sediment concentration near the bed are mainly sediment characteristics, bed shear stress, and turbulence. Rijn and Leo (1984) developed an empirical equation for near-bed sediment concentration using field and experimental results based on bed shear stress and particle diameter of the sediments. This equation has been used in the present model, which is given as

$$C_a = 0.015 \frac{d_{50}}{\delta_b}\left[\frac{|\tau_b| - \tau_{\text{cr}}}{\tau_{\text{cr}}}\right]^{1.5} D_*^{-0.3} \tag{13}$$

where τ_b = bed shear stress; τ_{cr} = critical bed shear stress; and δ_b = the reference level given as $0.01H$. The expression for bed shear stress was adopted from Soulsby et al. (1993):

$$\tau_b = \rho_w C_D \overline{U}^2 \tag{14}$$

where $\overline{U}$ is the depth-averaged total velocity and the drag coefficient C_D is given by

$$C_D = \left(\frac{0.4}{1 + \ln\left(\frac{d_{50}}{12h}\right)}\right)^2 \tag{15}$$

The expression for critical bed shear stress is given by

$$\tau_{\text{cr}} = g^*(\rho_s - \rho_w)^* d_{50}{}^* \theta_{\text{cr}} \tag{16}$$

where ρ_s = density of sediment particles; and θ_{cr} is the critical Shield's parameter given by (Soulsby 1997)

$$\theta_{\text{cr}} = \frac{0.3}{1 + 1.2 D_*} + 0.055[1 - \exp(-0.02 D_*)] \tag{17}$$

The expression for shear velocity u_{*c} is given by

$$u_{*c} = \sqrt{\frac{\tau_b}{\rho_w}} \tag{18}$$

where ρ_w = density of water.

Morphological Model

The morphodynamics involves the change in the shape of the bed in the river, sea, or lake due to the mass transport of sediments. The complex nonlinear waves generated by marine vessels churn out the bed sediments and ignite the sediment in suspension. Therefore, bed load and suspended load play a pivotal role in the bed evolution of the river morphology. The primary motivation in morphological modeling is to identify the areas of erosion and deposition in the channel due to vessel movement.

The modeling of bed morphology is accomplished by a one-dimensional Exner equation, which is given by

$$\frac{dz_b}{dt} = \frac{1}{1-n}[E - D - \nabla \cdot q_b] \tag{19}$$

where the erosion and deposition rate is captured by the E and D terms. Here, z_b is the time-dependent bed level change, n is the sediment porosity, and q_b is the bed load flux. The bedload flux is calculated by the Meyer-Peter and Müller (1948) formula given by

$$q_b = \frac{8[(\tau_b - \tau_{\text{cr}})/\rho_w]^{3/2}}{g(s-1)} \tag{20}$$

The above sediment continuity equation is used in the present model to understand the effect of sediment resuspension on the bed morphology. The mean sediment diameter used in our study was 0.1 mm, well below the sediment size mentioned by Meyer-Peter and Müller (1948). However, it was found from the

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literature that this formula works well for small-size sediment particles (Rijn and Leo 1984).

Validation of the Numerical Model

The numerical model is validated against the laboratory experiments of solitary waves in a wave tank (Kobayashi and Lawrence 2004). The experiment was conducted in a wave tank of dimensions 30 m long, 2.4 m wide, and 1.5 m deep. The experiment was performed on a sloping beach with an initial slope of 1 in 12 with breaking solitary waves. The waves were generated using a piston wavemaker with a depth of 0.8 m at the inlet. The sand particle used in the experiment had a d_{50} value of 0.18 mm, a porosity of 0.4, a specific gravity of 2.6, and a fall velocity w_s of 2 cm/s. Therefore, these sediment characteristics were used as the input in the numerical model. The data were recorded at eight gauge points. The first gauge was at the toe of the slope and the other gauges were located at a distance of 2, 3.5, 4.5, 6.5, 8.5, 9.6, 10.6 and 11.6 m from the first gauge. The initial wave height was 0.216 m. The setup is illustrated in Fig. 1.

Considering the dimension of the flume used in the experimental setup of Kobayashi and Lawrence (2004), we used a grid size $\Delta x = 25$ mm for our model simulation. Even though a 2D evaluation was carried out in the unidirectional flume, Kobayashi and Lawrence (2004) have not reported the erosion or deposition in the y direction. Hence, we have assumed this is a one-dimensional (1D) problem. So, the number of grid points in the y direction has been reduced. The grid points in the x and y directions are 1,240 and 6, respectively. The solitary waves of amplitude 0.216 m were generated using an internal wavemaker located at $x = 5$ m for $t = 0$ s. The water depth at the inlet was 0.8 m, with a sponge layer of 1 m placed at the west boundary. The bottom friction was added in the sediment transport model using a drag coefficient as given in Eq. (15).

The experimental results during the fourth solitary wave were used to validate the hydrodynamic model of FUNWAVE-TVD. The bed morphology change after four solitary waves is compared with the experimental data and FUNWAVE sediment transport module. The depth-averaged velocities obtained from the FUNWAVE-TVD were given as input into the sediment transport model at every time step. The sediment transport model computes the sediment flux, evaluates the erosion and deposition flux, and computes the bed evolution. The depth profile is thus updated after calculating bed evolution at every time step.

During the fourth solitary wave, the hydrodynamic model was validated against the experimental measurement at all the gauges. It can be seen from Fig. 2 that the model data are in good agreement with the experimental measurements. The hydrodynamic model well captures the solitary waves. The model rightly simulates the magnitude of the initial wave height.

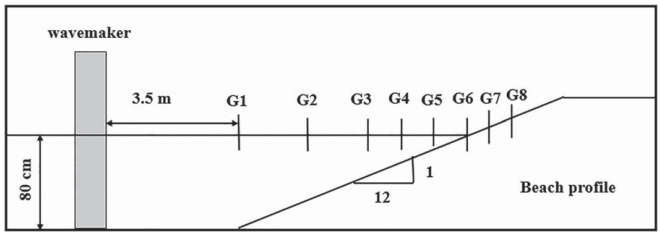


Fig. 1. Experimental setup of Kobayashi and Lawrence (2004).

The sediment transport and morphological model has been validated by comparing the beach profile after four solitary waves, as represented in Fig. 3. Based on finite volume, the inbuilt FUNWAVE sediment transport model was also adopted and reported in Fig. 3. The figure shows good agreement between the experimental measurements and the present sediment model compared with the existing sediment module in FUNWAVE. It is seen that erosion occurs in the shoreline region, mainly due to wave rundown, and is deposited in the upstream region. A similar phenomenon was observed in the experiment. The present sediment transport model shows a better comparison with the experimental data because of the modifications made in the FUNWAVE sediment transport module. The performance of the model in predicting erosion and deposition is listed in Table 1.

From Fig. 3, it is evident that the modified sediment transport model captures the overall shoreline profile well.

Evaluation of the Numerical Model Based on the Field Measurements

Proper validation of the FUNWAVE model for the moving vessel has not been reported to the best of the authors' knowledge. The ship-generated waves are validated against the laboratory experiments of Gourlay (2001). The work deals only with investigating turbulent bores generated by vessels traveling at supercritical speeds in a towing tank. However, a recent study in the Florida Intracoastal waterway by Forlini et al. (2021) reported that FUNWAVE-TVD shows excellent performance across all wake classes for weakly dispersive wakes. To evaluate the performance of the hydrodynamic and modified sediment transport model in restricted waterways, the simulated results of the numerical model are compared with the field-measured data of the ship wakes in the Falta stretch of the Hooghly river in the Indian National Waterway 1 (NW 1). We conducted the field study for 11 days, starting from August 23, 2021 and ending on September 2, 2021. The field measurements were carried out in the Falta jetty (22°15′17.57″N and 88° 5′16.19″E) on the Hooghly river, West Bengal, India. The continuous variation in the water surface elevation was measured using an ultrasonic wave-tide recorder (LOG aLevel, General Acoustics), which was extended 1 m outside the jetty to capture the ship waves. The river currents and backscattering data were collected using a horizontal-looking acoustic Doppler current profiler (ADCP). The ADCP was calibrated by performing a simple linear regression analysis of echo intensity (EI) obtained from the backscattered data of ADCP with the measured sediment samples at the Falta jetty. The bathymetry data were collected using Bathy 500 MF echo sounder and DGPS. The instruments were mounted in a launch vessel MV Sarthak and data were measured along with a predefined line profile across the river cross-section at an interval of 200 m during the high tide. The data were collected 2 km upstream and downstream from the Falta jetty. However, it was impossible to collect bathymetry data near the jetty and river banks due to insufficient depth and space for navigation of the launch vessel. Hence, only the sediment data were compared in the present study. More details about the field experiment campaign and data collection are reported by Chakraborty et al. (2022).

A computational domain of 1,632 m by 1,322 m was selected in the Falta stretch of the Hooghly river to simulate the ship-generated waves. The measured depth profile, along with the ship sailing path and location of the field measurement, are reported in Fig. 4(a). The depth profile reported in the figure is with respect to the Charted Datum (CD), which is 2.82 m below the mean sea level (MSL). The grid size of $\Delta x = 2.0$ and $\Delta y = 2.0$ m is used with a

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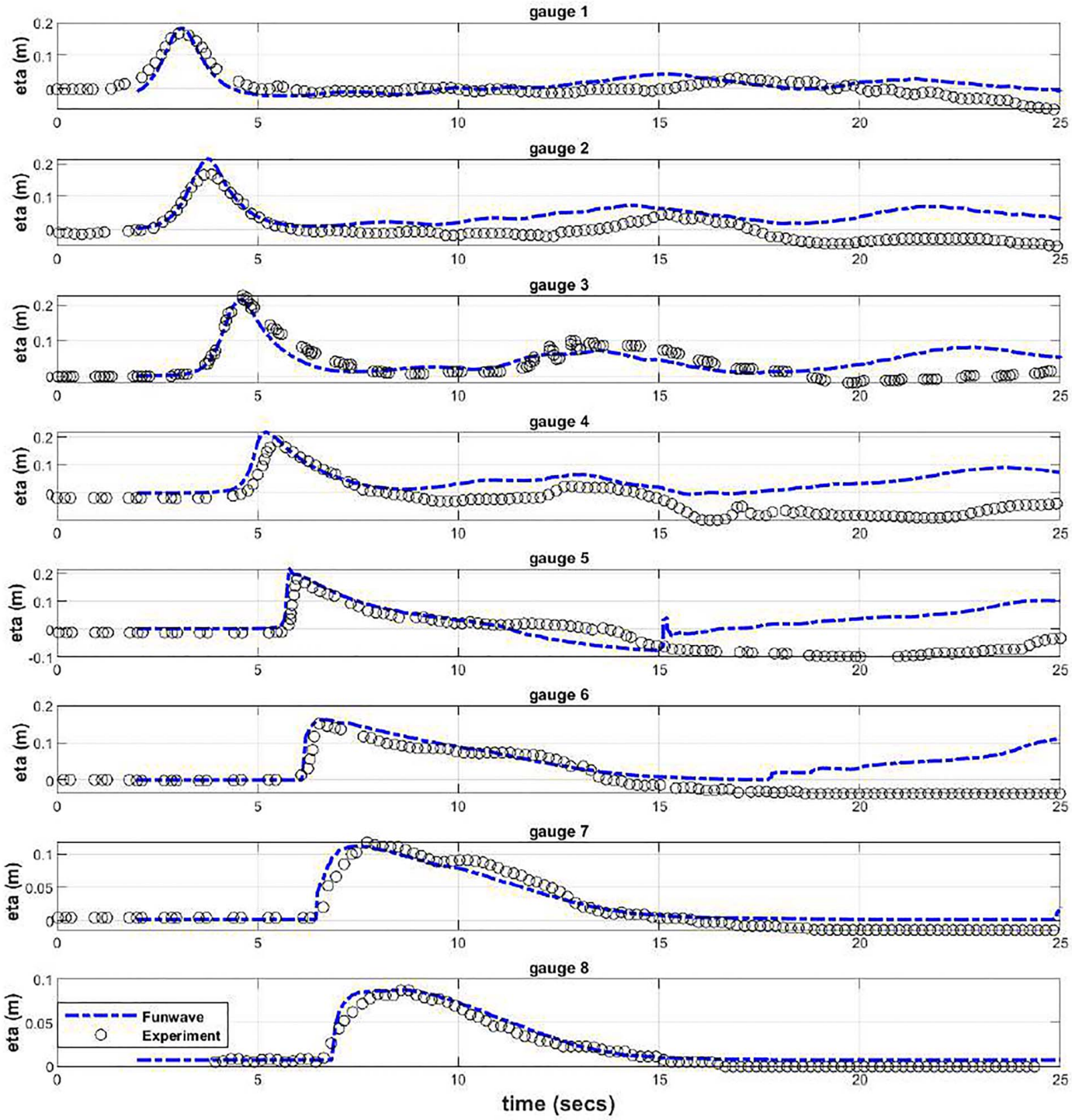


Fig. 2. Measured and simulated surface elevation during the fourth solitary wave at all the gauges listed in Fig. 1. The dash and dotted line represents the simulated data from the FUNWAVE and the open circle symbols represent the experimental data.

Courant–Friedrichs–Lewy (CFL) condition of 0.1. A sponge layer of 5 m is used in all the boundaries. The measured depth profile was converted to the same resolution of the computational domain by applying the linear interpolation function of MATLAB. The depth profile near the jetty was evaluated by averaging data from the neighboring grid points. The average sediment and water particle density in the Hooghly river are 2,650 and 1,025 kg/m^3, respectively. The d_{50} value of 1 mm and porosity of 0.4 is adapted in the present simulation. Ship SSL Sabarimalai has a length of 143.11 m, a beam of 22.8 m, and a draft of 7.0 m with block coefficients of 0.5, 0.5, and 0.5, as mentioned in Eqs. (5) and (6), which were deployed in the computational domain. The ship navigated through the sailing path represented in Fig. 4(a), with a constant speed of 5.92 m/s. The ship was placed at (1,000, 2,000 m) at time 0 s. The tidal elevation during the passage of ship SSL Sabarimalai was 6.3 m, which was added to the depth profile at the start of the simulation. The tide level was considered constant during the passage of the ship in the computational domain.

Fig. 4(b) represents a contour plot of the water surface elevation at 350 s. The significant wave height of the measured and modeled

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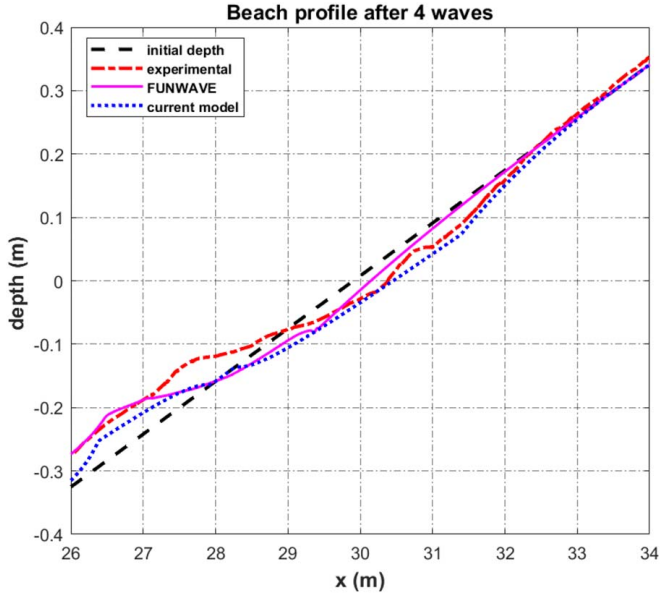


Fig. 3. Comparison of bed evolution between the present model and FUNWAVE-TVD sediment transport model against the experimental results after four solitary waves.

ship wake was 0.15 and 0.16 m, respectively. The energy content of the measured and modeled data was 0.2586 and 0.2542 J, respectively. Based on this statistical analysis, we could conclude that model results are consistent with measured data. Further application of the numerical model in Tallinn bay and Venice lagoon is reported in Appendices I and II.

Now, to evaluate the performance of the modified sediment transport module, the results of the depth-averaged sediment concentration (*sc*) from the model are compared with the results of the FUNWAVE sediment transport module as well as the measured *sc* data of the ship SSL Sabarimalai. The comparative plot of the *sc* data is in Fig. 5. It is clear from the figure that the modified sediment transport model shows a better prediction of the *sc* compared with the existing FUNWAVE sediment module. The maximum *sc* obtained from the analysis of the measured data was 4.3 mg/l. The maximum *sc* predicted by the modified sediment transport model and the FUNWAVE sediment transport model is 3.6 and 0.1 mg/l, respectively. The error value in predicting the *sc* by the FUNWAVE module is 97.6%, which is significantly reduced to 16.3% by the present modification. Therefore, it is concluded that the FUNWAVE hydrodynamic model and the modified sediment transport model can be used to simulate ship waves in the Hooghly river.

Table 1. Comparison between measured erosion and deposition with the present model data

Beach profile change	X (m)	Experiment (mm)	Funwave (mm)	Present Model (mm)
Deposition	27	59	55	34
	27.5	57	24	24
Erosion	30	27	7	27
	31	38	10	33

Parametric Study of Ship Wakes on Sediment Transport

Based on the validation of the numerical model with the field measurements for the moving vessel application, it has been applied to a typical restricted waterway in India. The government of India classified the inland waterways into 111 national waterways under the National Waterways Act 2016, out of which 13 national waterways are operational for shipping and navigation

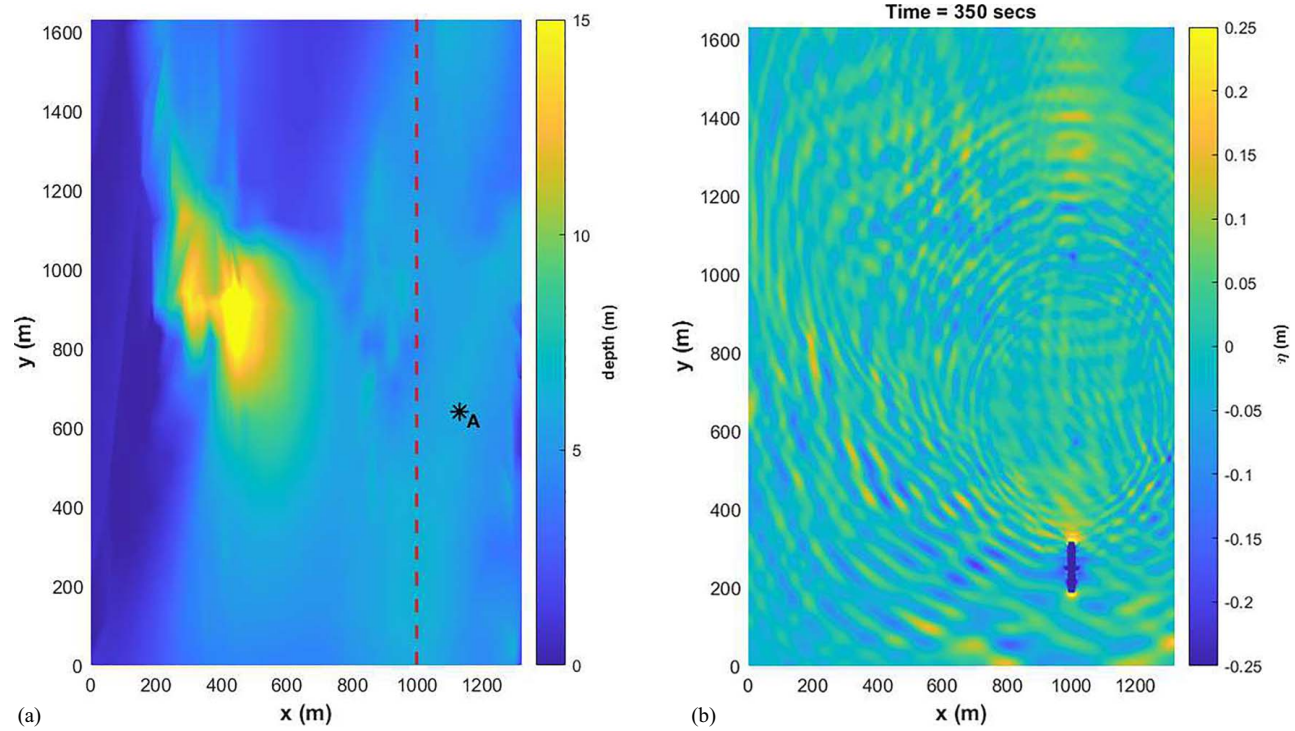


Fig. 4. (a) Contour plot of the depth profile in the computational domain. The dashed line represents the ship's sailing path. Point A is the location of the field-measured data; and (b) Snapshot of the water surface elevation profile of the ship SSL Sabarimalai at time 350 s.

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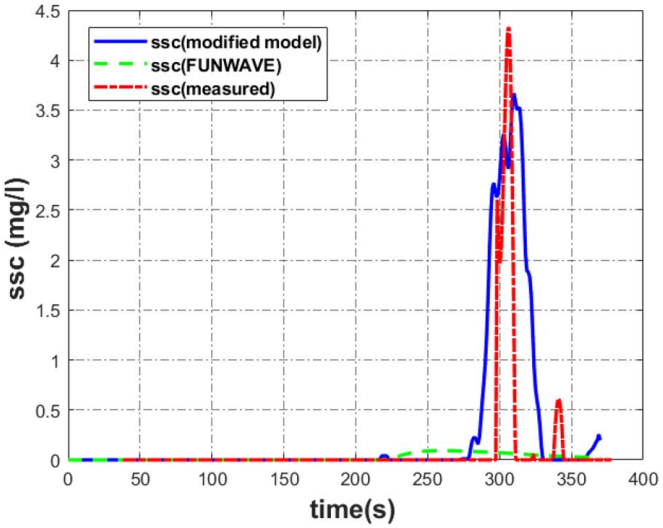


Fig. 5. Comparative plot of the *sc* from the modified sediment transport model, FUNWAVE sediment transport model, and measured data.

(Ministry of Ports, Shipping and Waterways, Government of India 2018). In 2019–2020, the number of high-draft vessels navigating the NW 1 was approximately 3,500. Hence, there is a possibility of a massive amount of sediment added to the channel due to marine traffic, which eventually increases the dredging cost of the navigational channel. The overview of the Hooghly river in NW 1 is represented in Fig. 6.

A typical study domain of 864×580 m has been considered with its origin at the left-hand corner. The grid size adopted was $\Delta x = 2.0$ and $\Delta y = 2.0$ m with a CFL condition of 0.1. A sponge layer of 5 m has been used.

The contour plot of the bathymetry used in the present study has been reported in Fig. 7(a). The representative bathymetry as that of the Uluberia stretch in NW 1 has been considered for the present



Fig. 6. Overview of the Hooghly river in NW 1, where point A denotes the Falta region and point B denotes the Uluberia region. (Image © Google, Image © 2023 Airbus, Image © 2023 Maxar Technologies.)

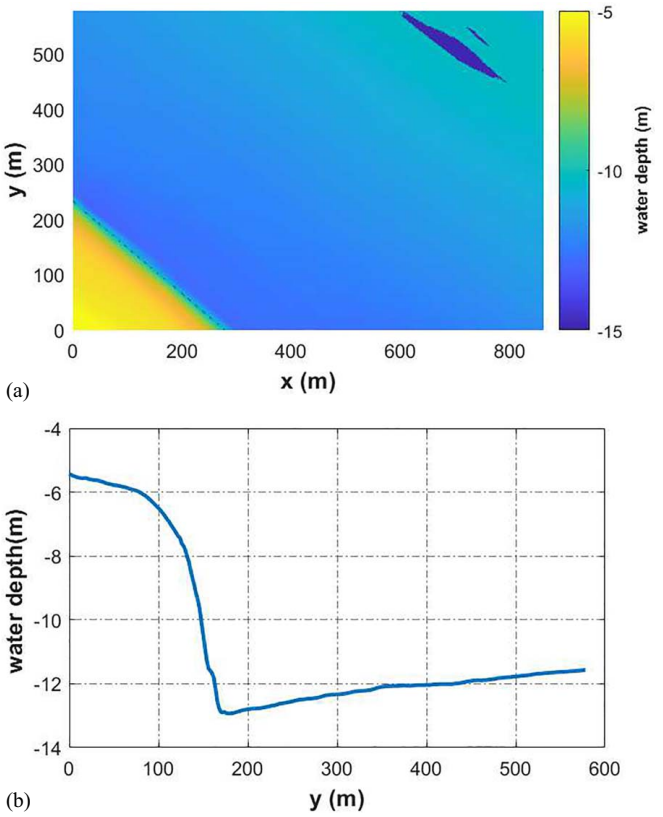


Fig. 7. Representation of the depth profile in the computational domain: (a) contour plot of bathymetry; and (b) cross section of the channel at transect $x = 100$ m.

study. The bathymetry measurements were obtained during the June 2019 field measurements. The channel depth varies from 5 m on the left side to 12 m on the right side [Fig. 7(b)]. The list of ships used in the current study is reported in Table 2. The ship speed and dimensions were obtained from MarineTraffic (2019). We chose this particular study domain in the Uluberia stretch to perform numerical experiments because the river width is narrow in this region compared with the Falta stretch. The sediment characteristics in this region are similar to that of the Falta stretch. Hence, the significance of the ship waves on this stretch is much more critical compared with the Falta stretch. The paper now focuses on studying the effect of these deep-draft vessels on sediment transport with the following parametric studies.

Simulation of Sediment Resuspension in the Channel

In this section, the ship Bon Atlantico has been used to understand the phenomenon of sediment resuspension. The ship was placed at 290 m in the computation domain. Then the ship was allowed to

Table 2. List of ships for numerical experiment

Ship Name	Ship Velocity V_{ship} (m/s)	Length L (m)	Breadth B (m)	Draft T (m)
Asiatic Moon	5.45	148.53	22.7	7.0
Xpress Yamuna	5.92	162.89	22.3	7.8
Bon Atlantico	6.17	134	23	7.2
Asiatic Neptune	6.43	148.53	22.7	6.1
Xpress Meghna	6.63	154.85	21.5	6.8
Hong Sen1	6.99	118	18.2	6.4

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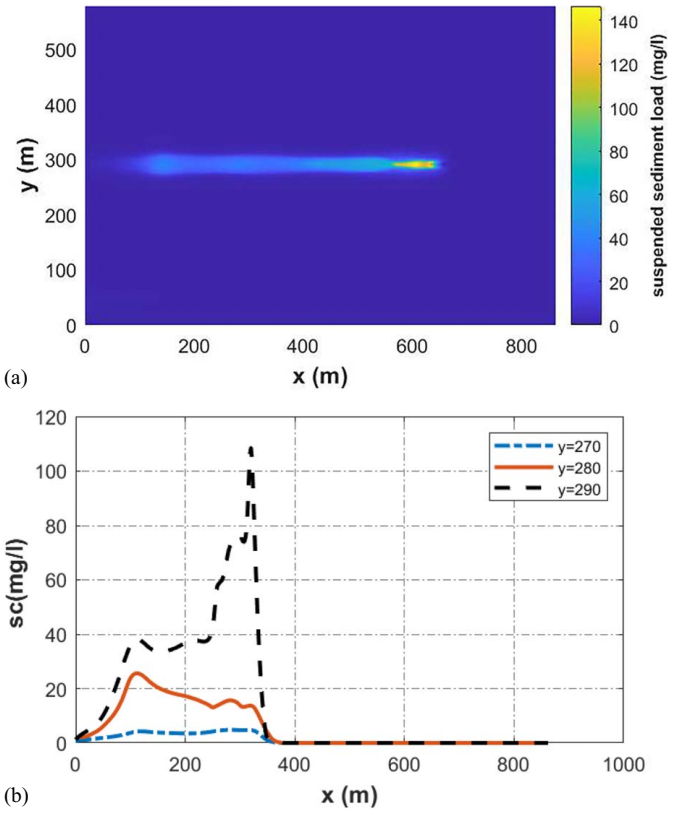


Fig. 8. (a) Snapshot of the sediment resuspension in the computational domain at $t = 100$ s for ship Bon Atlantico; and (b) Spatial distribution of depth-averaged sediment concentration at $t = 50$ s.

navigate at a constant speed of 6.17 m/s. The variation of the sediment resuspension in the restricted channel is listed in Fig. 8(a).

Fig. 8(a) gives a clear comparison between the patterns of sediment load movement due to the ship's passage in the domain. The primary and the secondary waves generated by the ship contribute to the initiation, resuspension, and distribution of suspended sediments in the channel. The resuspension of the sediments takes along the navigational path, front and back of the ship, as represented in Fig. 8(a). The study domain experiences a higher sediment concentration even after the ship's passage due to secondary waves.

The spatial distribution of the depth-averaged sediment concentration is represented in Fig. 8(b). The maximum value of 108 mg/l has been observed in the navigation path. As shown in the figure, there is a gradual decrease in sediment concentration as one moves from the navigation path to the banks due to surface wave dissipation. The two peaks in the figure at $y = 290$ m (navigation path) are mainly due to the ship's transverse waves generated by the bow and stern. Therefore, the transverse waves tend to cause greater sediment resuspension than the divergent waves.

Correlation of Froude Number with Ship Waves and Depth-averaged Sediment Concentration

As discussed above, the ship wake hydrodynamics in restricted waterways has been governed by two critical parameters: ship speed and water depth in the channel. The ship speed governs the generation of primary waves, secondary waves, and solitons for the deep-draft vessels in the channel. As the ship speed increases, the disturbance on both sides of the ship increases, causing suspension of sediments. To analyze the hydrodynamics and sediment concentration in restricted waterways, the depth Froude number (Fr) has been correlated with the maximum wave height (H_{max}) and maximum depth-averaged sediment concentration (sc_{max}). The maximum wave height has been calculated using zero up-crossing methods, that is, the vertical distance between the maximum trough and the corresponding maximum crest of the divergent wave group of the secondary waves (Chakraborty et al. 2022). The depth-averaged sediment concentration has been obtained by multiplying sediment density with the $\overline{c}$ obtained by solving Eq. (7). The point measurements of the H_{max} and sc_{max} were obtained at (200, 100 m) in the computational domain.

The ships generally navigate in sub-critical range in restricted waterways. The numerical simulation for depth Froude number varying from 0.25 to 0.7 was carried out for the ship Bon Atlantico. Fig. 9 represents the variation in the H_{max} and sc_{max} with Fr for varying speeds of Bon Atlantico denoted in circles. The H_{max} shows a linear increase with the increase in the depth Froude number as represented in Fig. 9(a). A similar trend was reported by Chakraborty et al. (2022), where the field measured H_{max} shows a steep increase with increasing Fr. The correlation study performed by Chakraborty

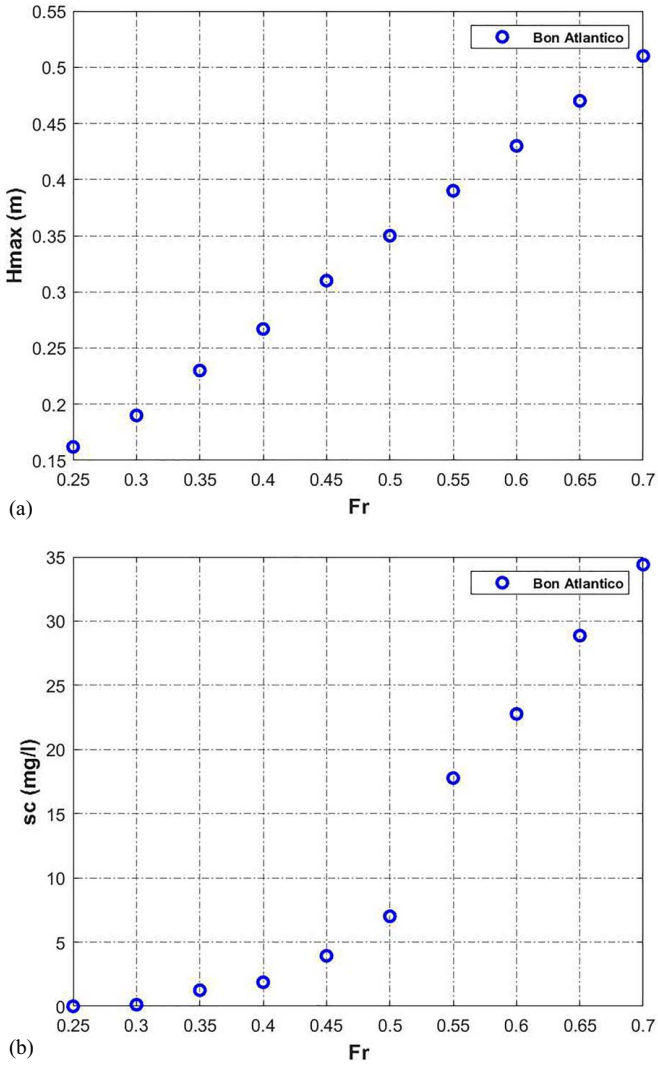


Fig. 9. (a) Variation of the maximum wave height generated due to ship Bon Atlantico in the channel is shown against the depth Froude number; and (b) Variation of maximum depth-averaged sediment concentration with the depth Froude Number. The point measurements were obtained at the location (200, 100 m) in the computational domain.

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et al. (2022) revealed that H_{max} was highly correlated with Fr and ship speed. Therefore, it can be concluded that the increase in the shipping speed in restricted waterways generates strong ship waves, which eventually cause river bed and bank erosion. This may jeopardize the sustainability of the waterway.

Similarly, in Fig. 9(b), it has been observed that the maximum depth-averaged sediment concentration increases with the increase in the depth Froude number. In contrast to the observation of H_{max}, the sediment concentration shows a nonlinear increase with the increasing Fr. The sediment concentration follows a steep increase for Fr ranging between 0.25 and 0.7 due to an increase in ship speed. In the following sections, different scenarios of the ship-wave–sediment interactions are simulated.

Scenario 1: Impact on the Sediment Transport due to Movement of Two Vehicles in Close Proximity

The restricted waterways often face a critical situation when two ships pass each other from opposite directions. Due to continuous operations, the ships will move in and out of the port. This section discusses the impact of sediment transport and bed morphology due to these conditions. Two ships, Bon Atlantico and Asiatic Neptune, have been used in this analysis. They are made to cross each other in the computational domain with a clearing distance of 100 m from the centerline of each ship. This is a hypothetical situation, wherein at the beginning of the simulation, the Bon Atlantico ship starts from (0, 240 m) and Asiatic Neptune starts from (1,000, 340 m).

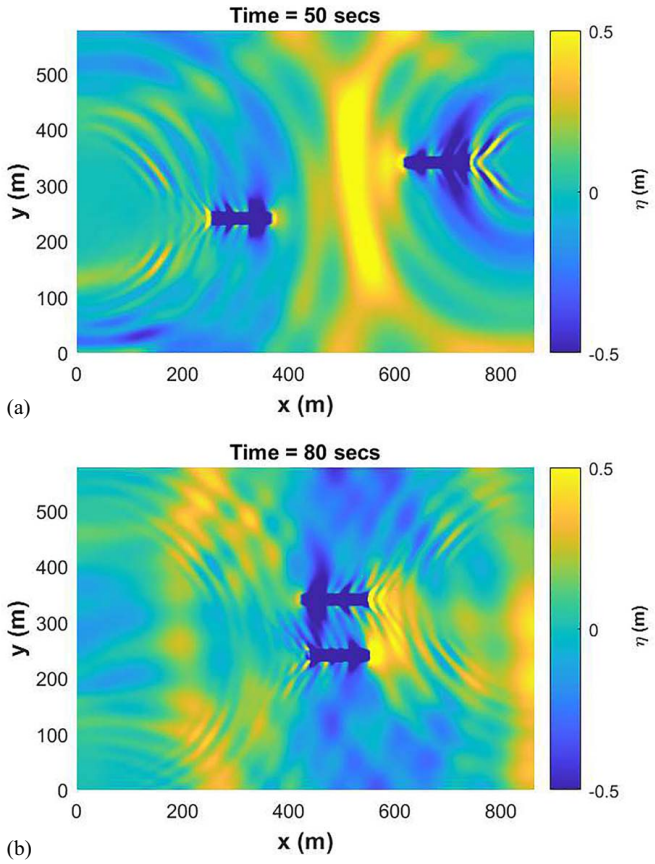


Fig. 10. Variation of surface elevation in the channel is shown due to the movement of Bon Atlantico and Asiatic Neptune separated by 100 m from the centerline of each ship: (a) contour plot of η at $t = 50$ s; and (b) contour plot of η at $t = 80$ s.

From Fig. 10(a), one can observe that in the range of 400 to 600 m, the ship bores (undular) generated by the two ships overlap each other and increase in magnitude, creating a massive disturbance in the channel. A comparison between the magnitude of the surface elevation of the two ships with respect to one ship's movement is in Fig. 11. The surface elevation increases by almost 2.5 times due to the interaction of the two ships. Similarly, the rise in ship bores creates a significant pressure variation in the channel, resulting in a big drawdown. This was noticed in the region between 600 and 800 m. The maximum drawdown of 0.32 m has been observed when the ship Bon Atlantico is moving alone in the path. However, the maximum drawdown significantly increases to 0.62 m when two ships cross each other. The most critical region may be when the bow section of the two ships is in line, as represented in Fig. 10(b).

The undular bores from the ship have a negligible effect on the sediment resuspension and bed morphology. Therefore, the channel does not experience any surge in sediment resuspension and bed erosion in the spatial ranges between 400 and 600 m. Fig. 12(a) reveals a steep increase in the sc in the region between 600 and 800 m due to the generation of drawdown and secondary waves. The sc increased from 0 to 1.5 mg/l at $x = 800$ m. Similarly, the domain experiences an increase in bed erosion at 800 m locations when the two ships pass. However, there is no erosion at the 800 m location when only a single ship navigates. This observation is clear from Fig. 12(b). Thus, it can be inferred that the passage of two ships in the restricted waterway creates a complex wave interaction, increasing the bed erosion and depth-averaged sediment concentration. Therefore, proper planning of ship navigation in restricted waterways is needed to reduce the increased sediment concentration. Otherwise, this will move with the currents and get deposited downstream, causing increased dredging operation.

Scenario 2: Net Sediment Flow in the Navigation Path due to Vessel Movement in the Opposite Direction

In the above section, we have discussed the effect of sediment transport due to the nearby movement of two ships. In restricted waterways, vessels often propagate in the same navigation path after crossing each other. This phenomenon happens before the calm water condition has been reached. Numerical simulations have been carried out for the ship Asiatic Neptune in the same navigation path from west to east after the ship Bon Atlantico navigated from east to west. This has been carried out assuming that the updated bathymetry has no suspended sediments in motion.

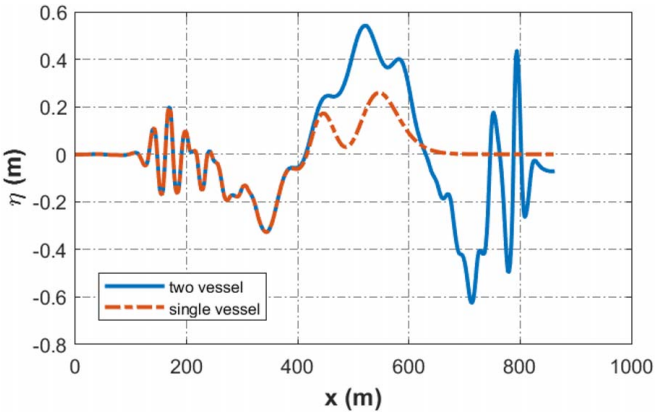


Fig. 11. Comparison plot of the spatial distribution of the surface elevation (at $y = 300$ m) of two ships against one ship at time 50 s.

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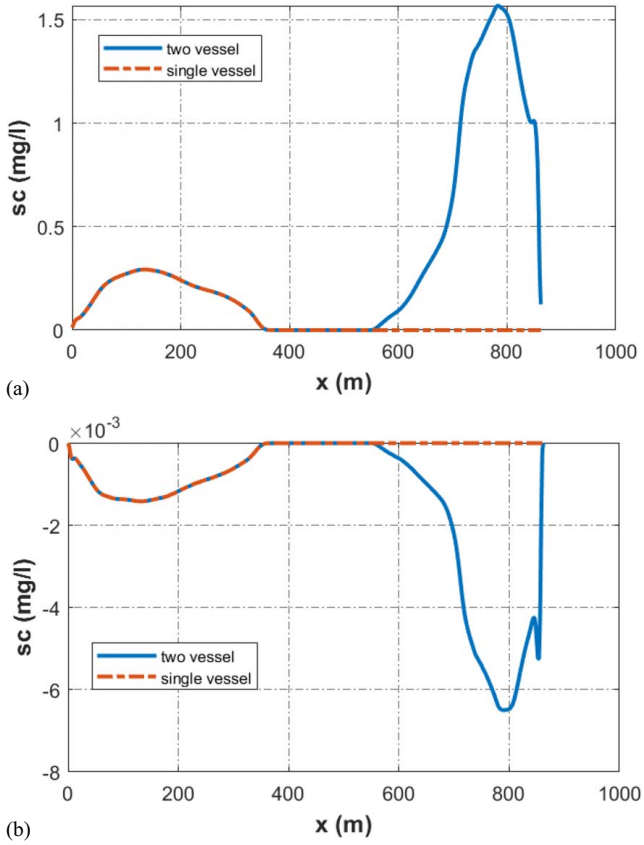


Fig. 12. Increasing effect on the (a) sediment transport; and (b) bed morphology plotted along the length of the domain at $y = 300$ m due to passage of two ships against one ship at time 50 s.

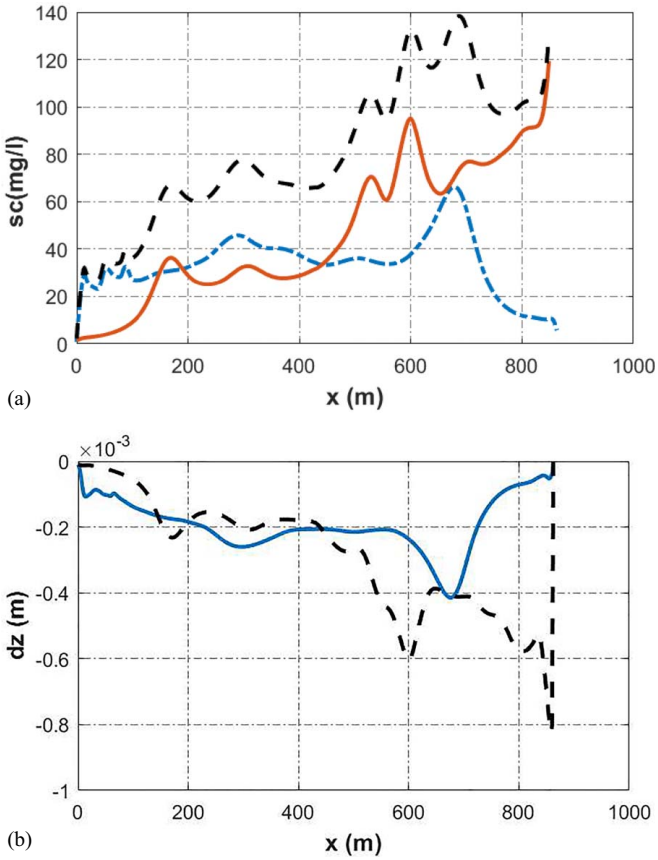


Fig. 13. (a) Spatial distribution of *sc* on the navigation channel after the passage of the ship. The dash and dotted line is the *sc* after the passage of the ship Bon Atlantico from east to west direction. The solid line is the *sc* after the passage of the ship Asiatic Neptune from west to east direction. The dashed line is the net sediment concentration in the navigation path. The net sediment concentration is obtained by the summation of the sediment induced by the passage of two ships; and (b) Spatial distribution of bed change on the navigation channel after the passage of the ship. The solid line is the bed level change (*dz*) after the passage of the ship Bon Atlantico from east to west direction. The dashed line is the *dz* in the navigation path after the passage of the ship Asiatic Neptune from west to east direction.

Thus, the updated bathymetry due to the sediment movement has been given as the input for the subsequent simulations.

Fig. 13(a) clearly represents that the resuspension of sediments is highly increased when two ships navigate through the same path in close time intervals coming from the opposite direction. The maximum *sc* generated by the individual vessels is 70 and 100 mg/l. The net sediment concentration is increased to 140 mg/l in the navigation path. The net sediment concentration is obtained by the summation of the sediment induced by the passage of two ships. The movement of the ship Bon Atlantico started the initiation of fine sediments from the bed. The immediate movement of the next ship Asiatic Neptune further increases the sediment resuspension. During the passage of a single vessel, the sediment concentration only increases for a short period and then decreases. This highly correlates with wave dissipation as it propagates. The movement of the second vessel, Asiatic Neptune, during this short period may enhance the sediment resuspension, which eventually may increase the river's turbidity and creates problems for aquatic life. The sediments thus suspended may flow with the river currents and get deposited in the downstream region, resulting in local morphological changes as reported in Fig. 13(b). Thus, it can be noticed from this hypothetical analysis that the net erosion in the bed may increase by two times.

Scenario 3: Impact of Multiple Vessel Movement on the Bed Morphology

Generally, based on past experiences in a particular channel and port operations, vessels pass through the waterways at an interval of five minutes. Therefore, the frequency of vessel movement plays a pivotal role in the bed evolution of the channel. To address these conditions, the ships are navigated in the same order listed in Table 2 in the computational domain. The total run time for each vessel was five minutes. Similar to scenario 2, the updated bathymetry was given as the input in the subsequent simulations.

The numerical experiments show a steep increase in bed erosion with the passage of each vessel. The bed erosion was evaluated at the location (200, 100 m) in the computational domain. In Fig. 14, the diamond indicates the erosion of the bed when the vessel moves individually in the domain. The black circle indicates erosion in the bed due to vessel movement one after another at an interval of five minutes. It is evident from the figure that bed erosion is highly accelerated due to the frequent passage of vessels in the domain. For this hypothetical simulation, one can see that the bed erosion after the last vessel's passage is 34 times higher than the initial vessel.

Scenario 4: Impact of the Small-Draft Vessel on the River Channel with Respect to Big-Draft Vessel

In the previous scenarios, we discussed various impacts of the deep-draft vessels in restricted waterways. However, studying

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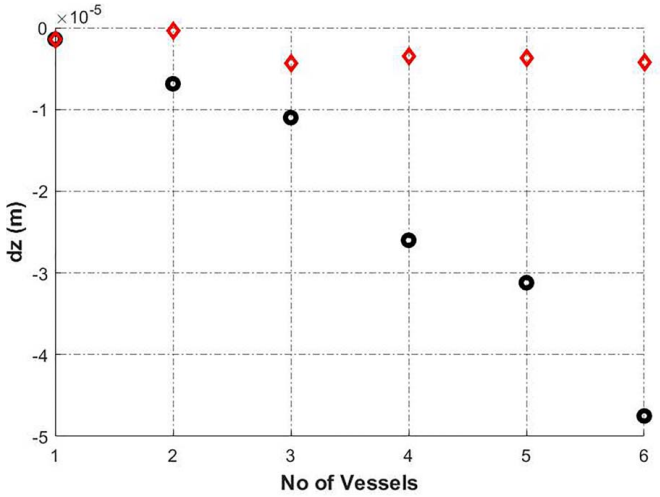


Fig. 14. Bed evolution change as a function of the frequency of vessels entering the computational domain. The diamond symbols denote the movement of the individual ships in the study area in the same order as listed in Table 2. The open circle symbols are the movement of the same ships in the same order, one after another, at an interval of 5 min.

the impact of the small-draft vessels in the channel is essential. The traffic of small-draft vessels is usually higher than that of higher draft vessels as there is no depth constraint in the channel. As discussed and observed in the previous sections, the impact of a single

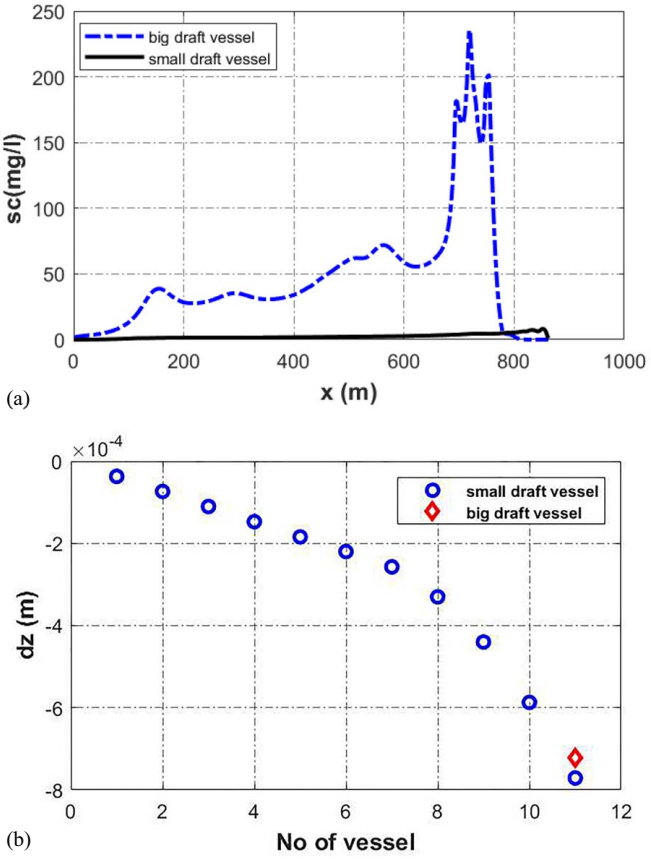


Fig. 15. (a) Comparative plot of the depth-averaged sediment concentration of ship Bon Atlantico to the small-draft vessel (Beki) at $y = 290$ m; and (b) Comparative plot of bed evolution of 11 small-draft vessels to the bed evolution of ship Bon Atlantico.

vessel in the channel is not of much importance. So, the numerical experiments considering a small-draft vessel (Beki) with dimensions of $81 \times 15 \times 2.1$ m that navigates through this channel with a speed of 7.5 m/s have been investigated. The same computational domain as used in the previous section has been considered. The comparison of the sediment concentration of the higher draft vessel with the small-draft vessel at $y = 290$ m is represented in Fig. 15(a). The numerical results reveal that the influence of the small-draft vessel is mainly in the vessel travel path and less than that of the big-draft vessel. Further, it indicates that the amount of sediment generated by the ship Bon Atlantico is almost 32 times higher than that of the small-draft vessel.

This result helps us to identify that the impact of a single small-draft vessel may not be influential in the river. However, the cumulative effect of many small drafts will be similar to the impact of a single big-draft vessel. To test this hypothesis, the simulations were carried out with 11 small-draft vessels, similar to scenario 3. It can be seen from Fig. 15(b) that the bed erosion (evaluated at the location (200, 100 m)) after 11 small-draft vessels is equal to the erosion caused by the passage of ship Bon Atlantico. Thus, as the number of small-draft vessels navigating in the channel is more than the deep-draft vessels, the cumulative impact of the small-draft vessel on sediment transport and bed evolution should not be ignored.

Summary and Conclusion

The main aim of the present study is to understand the impact of ship-generated waves in the complex hydrodynamics and sediment resuspension in restricted waterways. The numerical model FUNWAVE-TVD with a modified sediment transport model has been adopted. The current sediment transport model used different diffusivity coefficients related to the turbulent eddy viscosity. The expression for deposition fluxes used in the current model produced comprehensive results. The current model incorporates the user-defined background concentration. The present model used an explicit expression for fall velocity and shield parameter. The modified numerical model was validated against the experimental results of Kobayashi and Lawrence (2004). The hydrodynamic model showed good agreement with the experimental measurements at all eight gauge points. The modified sediment transport model captured the changes in the shoreline profile after four solitary waves. The prediction of erosion and deposition was significantly increased by using the modified sediment transport model. The implementation of the moving vessel in the numerical model and its impact on suspended sediment concentration was evaluated against the field measurements in the Falta stretch of the Hooghly river in Indian National Waterway 1. The error in predicting the depth-averaged suspended sediment concentration by the existing FUNWAVE sediment transport module is 97.6%, which is significantly reduced to 16.3% based on the present modification.

The vessels mainly navigate in the sub-critical range in restricted waterways, leading to the generation of primary and secondary waves. The primary waves mainly govern the initiation of sediments. The secondary waves caused the accelerating effect on sediment resuspension, advection, and diffusion of sediments in the channel. Another important factor in the initiation of sediments is the propeller influence in deep-draft vessels. The present 2D depth-averaged numerical model has certain limitations in incorporating the jet flows. The propeller-induced sediment resuspension is an important study and contributes substantially. This is one of the limitations of the present study. Nevertheless, the notable outcome of the present study is listed as follows.

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- The concentration of the sediment gradually decreases from the navigation channel to the river banks due to dissipation in the ship waves. It was found that the vessel speed and the river bathymetry are the critical parameters in the generation of the secondary waves, resuspension of sediments, and change in bed morphology in the channel.
- The morphological model revealed that the bed load movement is a local phenomenon and has less impact on the channel dynamics than the suspended load.
- The analysis of the maximum wave height with the depth Froude number reveals that H_{max} increases with the increase in Fr and ship speed, whereas sediment concentration shows a nonlinearity.
- The hypothetical situation considering the interaction of two ships from opposite directions increases the surface elevation by 2.5 times. Under these conditions, the sediment concentration is almost five times the usual.
- In the hypothetical condition, where the Asiatic Neptune ship crosses the navigation path within a short period after the ship Bon Atlantico, the net sediment flow in the channel increases to 140 from 70 mg/l.
- The present study reported that the bed erosion increases by 34 times when the ships are navigated in a short interval of 5 min against individual movement. Therefore, the time interval of vessel movement is an important parameter that causes the stimulating effect of sediment resuspension. In practical situations, enhanced sediment resuspension may occur due to a lack of adequate time in river bed compaction after the movement of the first vessel.
- The hypothetical comparative analysis of the small-draft vessel (Beki) with the big-draft vessel (Bon Atlantico) revealed that the sediment resuspension due to the big-draft vessel was 32 times higher than the small-draft vessel.
- Similarly, it was reported that the cumulative effect of the 11 small-draft vessels matched the channel morphodynamics of a big-draft vessel. Therefore, the cumulative effect of the small-draft vessels substantially affects the river morphology and cannot be neglected.

From the present study, it can be concluded that the limitation of the vessel speed (less than 5 m/s) and the increase in the time interval of ship movement (more than 5 min) will substantially reduce sediment resuspension due to ship-generated waves. Navigating the entire ship traffic in a predefined path leads to critical situations, which causes an abrupt change in the river bathymetry. Therefore, the numerical model may be applied to the national waterways to understand the effect of the abovementioned critical situations and establish an effective navigation plan.

Appendix I. Validation of Moving Vessel in the Hydrodynamic Model

The implementation of the moving vessel in the hydrodynamic model has been validated against the field measurements at Tallinn Bay (Torsvik et al. 2009). The proper implementation in Tallinn Bay could justify the model's overall performance in restricted waterways. The measurements were reported on the southwest coast of Aegna as listed in Fig. 16 from June to July, 2008. The water surface elevation due to ship wakes was recorded using an ultrasonic wave-tide recorder (LOG aLevel). The instrument was placed 100 m from the coast, outside the main shoaling region and surf zone. Further details about the measurement program have been reported by Parnell and Kofoed-Hansen (2001).

A computational domain of 4.5 by 7.5 km was selected to simulate the ship wakes from the site location. The measurement site location was sheltered from the southbound tracks, so in the present paper, the measurements from the northbound track are used for better wave conditions. The FUNWAVE model was validated from the measurement of track 1 at location 4 [WG 4, Fig. 16(a)]. Due to inaccuracies in the digitized topography, the water depth at location 4

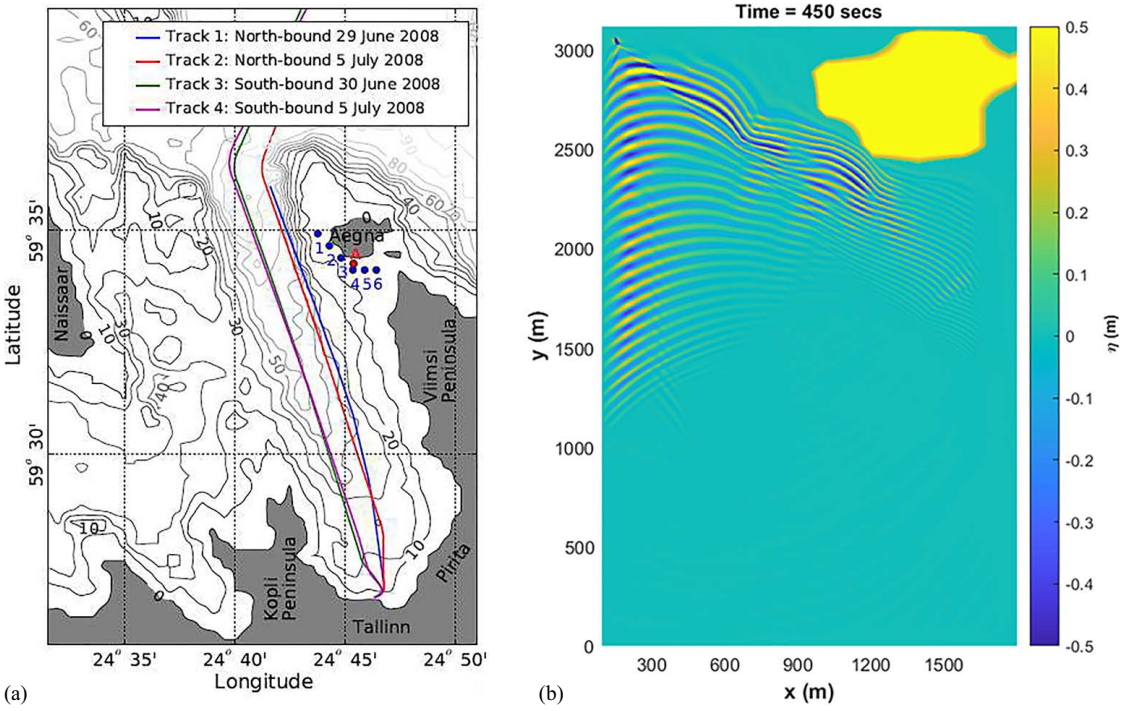


Fig. 16. (a) Location of the field survey at Tallinn Bay along with ship track and measured data location, collected from Torsvik et al. (2009); and (b) Snapshot of water surface elevation at time 450 s at northbound track 1.

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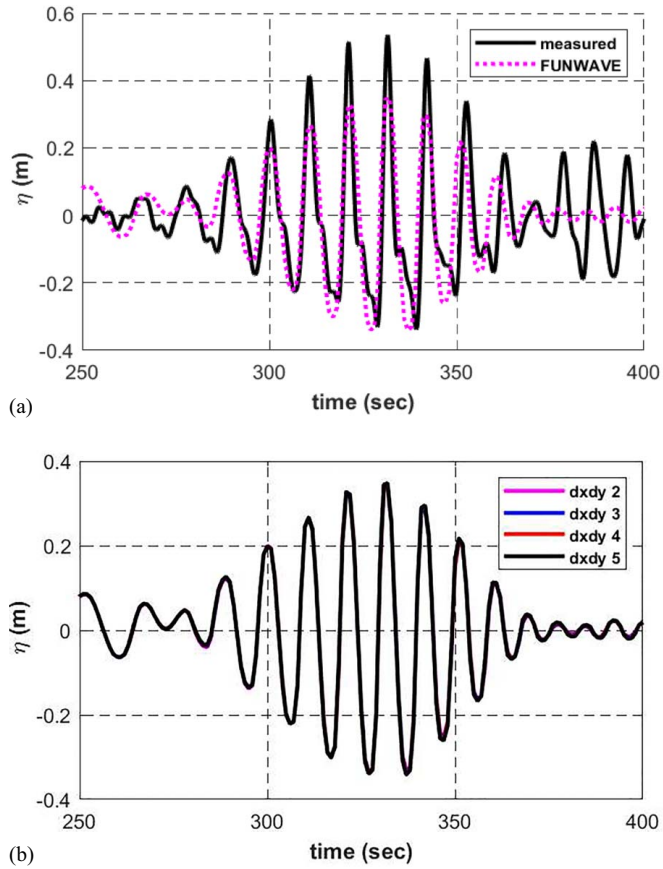


Fig. 17. (a) Measured and simulated wave elevation for track 1 at WG 4; and (b) Convergence study on the hydrodynamic model in Tallinn Bay.

was 2.7 m. The water depth in the location was observed to be 3.9 m from the field experiments. So, this wave gauge is more representative of waves outside the shoaling zone. In the FUNWAVE model, a structured grid of 5 by 5 m is considered to simulate the ship wakes by adopting the Courant–Friedrichs–Lewy (CFL) condition of 0.5. As represented in Fig. 17(b), a convergence study was performed. The model works fine for various grid sizes. A ship of length 100 m, width 40 m, and block coefficients of 0.75, 0.75, and 0.75, as mentioned in Eqs. (5) and (6), were considered to navigate through track 1 at a constant speed of 15.95 m/s.

Fig. 16(b) gives a good representation of the ship wave propagation toward the island. The wave groups from the measurements have high amplitude initially and successive wave groups have small amplitude. The FUNWAVE model shows a similar pattern in the computation, with the higher amplitude waves being well captured and reaching the bay first. Fig. 17(a) represents that the measured and simulated data are in good agreement. The FUNWAVE model slightly under-predicted the variation in the water surface elevation due to inaccuracies in the bottom topography, as wave shoaling was not captured accurately. Therefore, the comparison of the water surface elevation after 400 s was discarded. Nevertheless, considering the field measurements and the various uncertainty, the FUNWAVE results are acceptable for process understanding.

Appendix II. Validation of Sediment Resuspension due to Ship Waves

After successfully implementing the moving vessel in the hydrodynamic model, the primary aim is to validate the present sediment transport model due to the moving vessel application. The model has been validated against the field measurement results in the Venice lagoon as reported by Gelinas et al. (2013). The field measurements were performed by Gelinas et al. (2013) in the Malamocco–Marghera Canal, with the Porto Marghera Industrial Zone (PMIZ) located on the western shore of the lagoon, as listed in Fig. 18. More details about the instruments and deployment locations are reported by Gelinas et al. (2013) and will not be repeated here. Further, the channel bed was sandy and composed of 56.3% sand, 27.53% silt, and 16.17% clay (Bellucci et al. 2002), and the mean bulk density in the lagoon bed near the study area was reported to be $1,907 \pm 60\ \mathrm{kg/m^3}$ (Amos et al. 2004). The dimensions of the ship traffic navigating in the lagoon have a length between 114 and 208 m, a beam between 9 and 32 m, and a draft varying between 4.5 and 9.1 m, which is similar to the ship traffic dimensions used in the subsequent sections. The vessels at Venice lagoon navigate in the sub-critical range. Therefore, the modified sediment transport model has been applied at the Venice lagoon. As discussed in subsequent sections, the overall channel dynamics and input parameters are homologous to the present study domain.

The background concentration reported in the paper varies between 30 and 100 mg/l due to strong winds. The paper reported that the sides of the channel were steep, rising toward the shoal with a slope of approximately 0.25 on the east side and 0.30 on the west side (Rapaglia et al. 2011). The channel depth was maintained at 12 m due to periodic dredging in the canal. The average water density in the lagoon is $1{,}025\ \mathrm{kg/m^3}$ (Ferrarin et al. 2010), which is given as the input in the model.

A computational domain of 1,000 m by 200 m, as in Fig. 18, was chosen to estimate the sediment resuspension due to ship waves at the location (200, 50 m) in the study domain. A constant depth of 12 m was used for computation as per the site condition. The

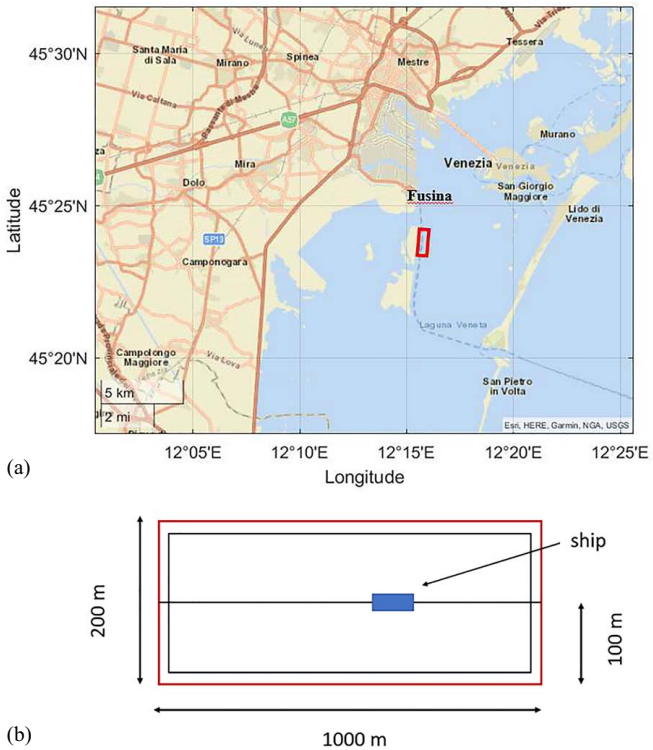


Fig. 18. (a) Study location of the measurement of the sediment resuspension due to ship waves in Venice Lagoon. The closed boundary shows the computational domain; and (b) Enlarged view of the computational domain. (Sources: Esri, HERE, Garmin, NGA, USGS.)

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Table 3. List of maximum suspended sediment concentration (sc_{max}) from the field measurement and sediment transport model

Ship Name	Field sc_{max} (mg/l)	Model sc_{max} (mg/l)	Direction
Grande Silicia	381.46	1772.10	Inbound
East Coast	33.41	33.0	Inbound
Clipper Karina	33.67	36.72	Inbound
Tucana	67.92	84.47	Outbound
Coral Leaf	302.93	271.5	Outbound
Jia Xing	83.68	68.56	Outbound
Ain Zeft	35.47	35.82	Outbound
MSC Mirella	81.66	80.14	Outbound
Novorossiysk Star	381.46	737.8	Outbound
Uni Assent	127.45	94.74	Outbound
Salerno Express	142.52	116.58	Outbound
M/T Tigullio	328.18	209.07	Outbound

grid size of $\Delta x = 2.0$ and $\Delta y = 2.0$ m is applied with a CFL condition of 0.1. The results of the maximum suspended sediment concentration (sc_{max}) from the sediment transport model against the field data are listed in Table 3.

Table 3 lists a considerable deviation between the measured and computed maximum suspended sediment concentration for ships Grande Silicia and Novorossiysk Star. As the two ships navigated through the study area, the sediments' resuspension exceeded the current meter's upper limit, which is 381.46 mg/l, and the actual suspended sediment concentration could not be measured. Further, the current meter did not capture any sediment resuspension when the ships East Coast, Clipper Karina, and Ain Zeft navigated through the study area (Gelinas et al. 2013). However, the numerical model showed zero sediment resuspension for these ships as the bed shear stress was less than the critical value. The comparative study between the analytical (Gelinas et al. 2013) and simulated (present model) bed shear stress and suspended sediment concentration is reported in Fig. 19. Figs. 19(a and b) represent that the model slightly under-predicted the bed shear stress because of the drag coefficient obtained from Eq. (15) is slightly less than the value used by Gelinas et al. (2013).

Fig. 19(c) represents that the model predicts the maximum suspended sediment concentration for the above data set except Grande Silicia and Novorossiysk Star. The slight deviation with the increasing sc_{max} values is because of the under-predicted bed shear stress values, which is acceptable. The comparative analysis of sediment resuspension, including ships Grande Silicia and Novorossiysk Star, is in Fig. 19(d). Therefore, the model addresses the limitations faced in the field measurements of the sediment resuspension due to moving vessels in the Venice lagoon. It should be noted that the implementation of the background concentration

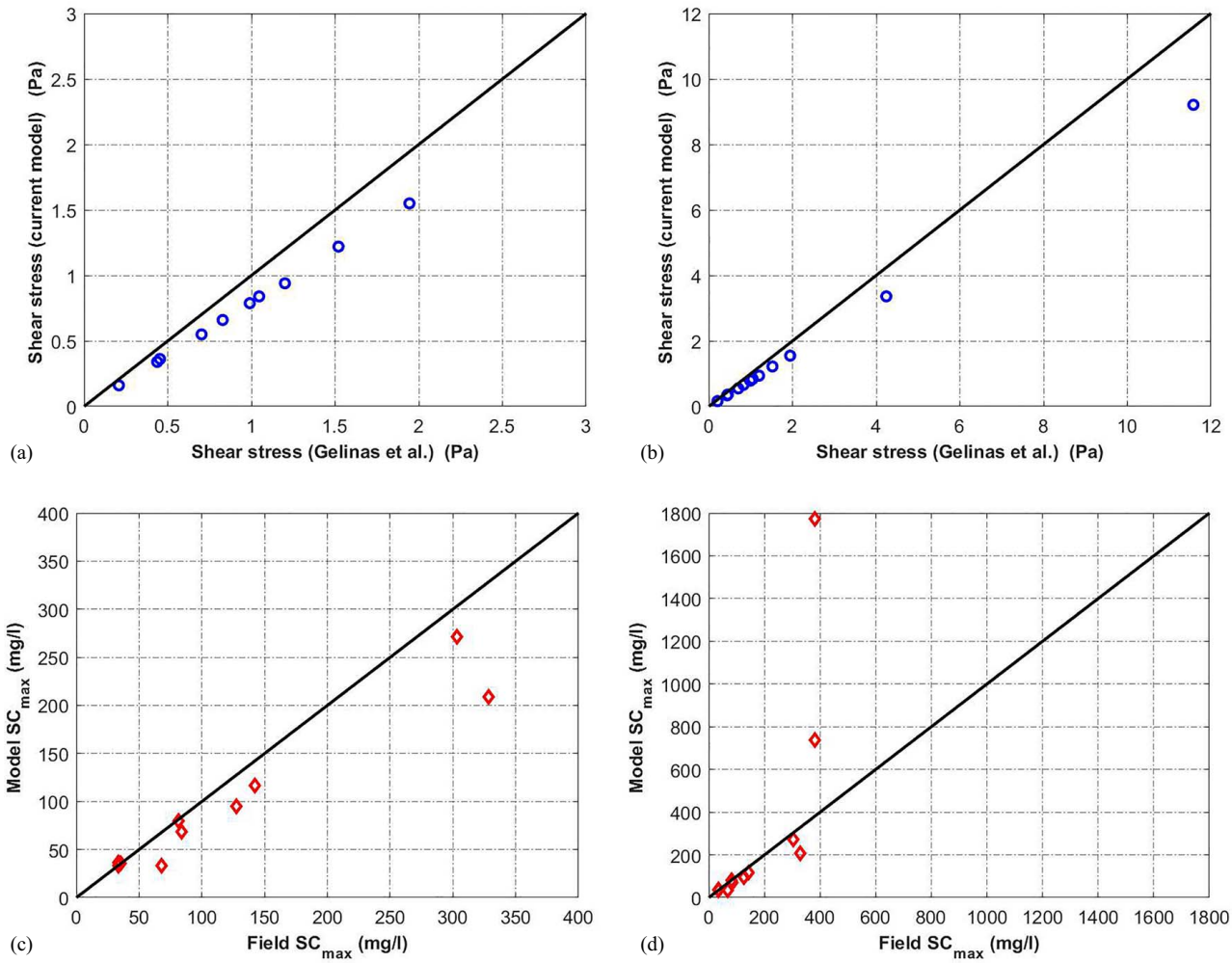


Fig. 19. (a) Comparison between analytical shear stress (Gelinas et al. 2013) and Model shear stress [as per Eq. (14)] without ship Grande Silicia and Novorossiysk Star; (b) with Grande Silicia and Novorossiysk Star; (c) Comparison between measured and simulated sc_{max} without ship Grande Silicia and Novorossiysk Star; and (d) with Grande Silicia and Novorossiysk Star.

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in the present sediment transport model helped to produce a better agreement with the field-measured data.

Data Availability Statement

Some of the data used during the study were provided by a third party. Direct requests for these materials may be made to the provider as indicated in the Acknowledgements. Some of the data that support the findings of this study are available from the corresponding author upon reasonable request. All other data generated or used during the study appear in the published article.

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